

Rahul Narayanan

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Industrus
Sydney, NSW

Dear Hiring Manager,

Re: Graduate Engineer Program

I am a final-year Mechatronic Engineering (Honours) student at the University of Technology Sydney, on track for First Class Honours in November 2026, applying for the Industrus Engineering Graduate Program. The program's mix of project rotations, structured development and multidisciplinary consulting work strongly matches the way I want to begin my engineering career. Through industry experience and university projects, I have developed practical experience across embedded software, robotics, requirements definition, systems integration, testing and multidisciplinary teamwork.

3.1 Ethical conduct and professional accountability

During my internship at Optik Consultancy, I worked on a respiratory-testing prototype for Royal North Shore Hospital. Because the usefulness of the system depended on reliable sensor acquisition, interface behaviour and recorded data, I treated testing and validation as a professional responsibility rather than a final check. I verified state transitions, sensor readings, data logging and end-to-end test sequences, raised integration issues when they appeared and iterated until the workflow operated reliably. This reinforced the importance of taking ownership of engineering decisions and their consequences.

3.2 Communication with engineers and stakeholders from different fields

My current work with Pavement Management Services requires me to translate industry needs into practical software and functional requirements for a modular rooftop camera system. I discuss camera, compute and interface constraints with software, electrical and mechanical team members so that decisions made in one area remain compatible with the wider design. At Optik, I also coordinated hardware-software interface requirements with a multidisciplinary team. These experiences have taught me to adjust the level of technical detail to the audience while keeping decisions clear and traceable.

3.3 Creative, innovative and proactive approach

I enjoy open-ended engineering problems where there is more than one workable solution. At Optik, our team helped transform the functionality of a non-portable AU\$70,000 respiratory-testing system into an accurate handheld prototype costing under AU\$300. In my capstone project, I am independently developing a Follow-Me Cooling Cart that combines sensing, computer vision, embedded computing and motion control. I also started Osprey, a C++ chess engine, to strengthen my ability to profile, benchmark and optimise complex software outside assessed coursework.

3.4 Ability to use and manage information

At Pavement Management Services, I research and evaluate NVIDIA Jetson platforms and camera hardware against processing requirements, resolution, frame rate, connectivity and software compatibility. I use that information to support hardware selection and convert findings into requirements and system-level decisions that other team members can work from. Across my projects, I also use test results, benchmarks and technical documentation to compare alternatives and refine designs instead of relying only on intuition.

3.5 Ability to manage my own performance in a professional environment

During my Optik internship, one challenge arose when different parts of the respiratory-testing prototype did not always behave as expected once they were integrated, even when individual functions had worked correctly on their own. Rather than treating each fault in isolation, I broke the problem into smaller testable areas covering the touchscreen interface, finite state machine logic, sensor acquisition and data logging. I compared expected and actual behaviour, isolated faults, discussed interface issues with the wider team and repeated the relevant tests after each change. This process allowed us to progressively stabilise the complete respiratory-test workflow and

reinforced the importance of responding methodically when an engineering system does not work as expected. I now apply the same approach by breaking larger problems into testable tasks, setting checkpoints, reviewing results and seeking feedback when required.

3.6 Teamwork and leadership

At Optik, I worked as Embedded Software Lead within a multidisciplinary engineering team, taking responsibility for the ESP32 firmware architecture while coordinating interfaces with the electrical and mechanical work around it. In a High Distinction wearable fitness tracker project at UTS, I also served as group leader while contributing to the embedded software and helping organise work across software, electrical and mechanical responsibilities. These experiences taught me that leadership is not about doing every task myself, but about keeping responsibilities clear, supporting communication and helping the team reach a reliable integrated result.

Industrus Engineering's exposure to varied projects, mentoring and professional development would give me the opportunity to apply these strengths while continuing to learn from engineers across different disciplines. I am available to commence full-time employment from January 2027 and would welcome the opportunity to contribute to the Graduate Program. Thank you for considering my application.

Kind regards,
Rahul Narayanan